MATHEMATICS – F2 – 2026 (MARCH)

MARKING SCHEME – PRE-MOCK EXAMINATION

ARUSHA REGION.

1 (a) (i) physics
(ii) Business studies
} Students to give two point to each subject. (04) marks

(b)(i) Salary = Tshs 250,000

→ One-fifth spent on fuel

$\frac{1}{5} \times 250,000 = 50,000$

→ Half of remaining Spent on Food

$\frac{1}{2} \times 200,000 = 100,000$

→ Half of the rest spent on miscellaneous (04) marks

$\frac{1}{2} \times 100,000 = 50,000$

∴ Monthly saving = Tshs 50,000

(ii) Yes, because 50,000 can be written as $\frac{50000}{1}$ (02) marks.

2 (a) Given

$A:B = 2:3$ ×5
$B:C = 5:6$ ×3

$A:B = 10:15$ 01

$B:C = 15:30$ 01

Hence $A:C = 10:30$ 01

or $A:C = 1:3$

2 (b) Rice : Wheat : Corn = 2 : 3 : 5

Mass of Rice = 3 kg

$2+3+5=10$ 01

Let x be the total mass of cereals

Then for Rice :

$$\frac{2}{10}x = 3$$

$$x = \frac{3\times 10}{2} = 15 \text{ kg}$$ 01

Mass of Corn $= \frac{1}{\cancel{10}_{2}} \times 15 = \frac{15}{2} = 7.5$ 01

$= 7.5$ kg 01

(c) Let $x = 0.5\dot{6}$ -- (i)

$10x = 5.6\dot{6}$ -- (ii)

(ii) − (i)

$10x - x = 5.6\dot{6} - 0.5\dot{6}$ 01

$\frac{9x}{9} = \frac{5.1 \times 10}{9 \times 10}$ 01

$x = \frac{51}{90} = \frac{17}{30}$

Hence $0.5\dot{6} = \frac{17}{30}$ 01

(a) $|2x+1| \leq 3$

$\pm(2x+1) \leq 3$

$+(2x+1) \leq 3$

$2x+1 \leq 3$

$2x \leq 3-1$

$\frac{2x}{2} \leq \frac{2}{2}$

$x \leq 1$ 02

$-(2x+1) \leq 3$

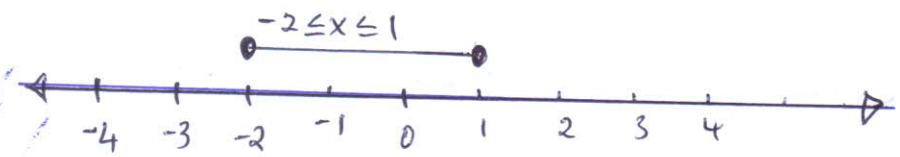
$-2x-1 \leq 3$

$-2x \leq 3+1$

$-2x \leq 4$

$\frac{-2x}{-2} \geq \frac{4}{-2}$

$x \geq -2$ 02



01

(b) (i) $6.645 \simeq 6.64$ since digits after 5 are all zeros 01/2

(ii) $8,125,600 \simeq 8,130,000$ since digit after 5 is greater than zero. 01/2

(c) $4x - 3(2x-5)$

$4x - 6x + 15$

$-2x + 15$ 02

4 (a) Complementary angles their sum is 90.

$A + B = 90$

$A = 2B$

$2B + B = 90$

$\frac{3B}{3} = \frac{90}{3}$

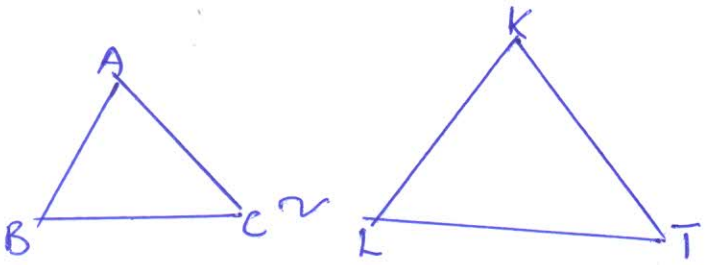
$B = 30$

$A = 2(30) = 60$

05 marks

Angles are 30° and 60°

(b)



(i) $\Delta ABC \sim \Delta KLT$ because their corresponding sides are proportional.

(ii) $A\hat{B}C$ correspond to $K\hat{L}T$

$B\hat{A}C$ —11— $L\hat{K}T$

$A\hat{C}B$ —11— $K\hat{T}L$

05 marks

5 (a) $\frac{x-6}{7} - \frac{2x+3}{4} = \frac{x-5}{2}$

$\frac{4(x-6)-7(2x+3)}{28} = \frac{x-5}{2}$

$\cancel{28} \times \frac{(4x-24-14x-21)}{\cancel{28}} = (x-5) \times \frac{14}{\cancel{2}}$

$-10x-45 = 14x-70$

$-45+70 = 14x+10x$

$\frac{25}{24} = \frac{24x}{24}$

05 marks

$x = \frac{25}{24}$

(b) $+\begin{cases} 2x+y=8 \\ x-y=1 \end{cases}$

$\frac{3x}{3} = \frac{9}{3}$

$x = 3$

$x-y=1$

$3-y=1$

$3-1=y$

$2=y$

05 marks.

$\therefore x=3$

$y=2$

6. (a) $4x + 5y = 40$

(i) $\frac{5y}{5} = -\frac{4x}{5} + \frac{40}{5}$

$y = -\frac{4}{5}x + 8$ 02

(ii) From $y = mx + c$

m is gradient

$y = -4/5\,x + 8$

$m = -4/5$ 02

gradient of the straight line $= -4/5$

(iii) x - intercept is when $y = 0$

$4x + 5(0) = 40$

$4x + 0 = 40$

$\frac{4x}{4} = \frac{40}{4}$

$x = 10$

$\therefore$ x - intercept $= 10$ 01

y-intercept $= c$ or when $x = 0$

$y = -\frac{4}{5}(0) + 8$

$y = 8$

$\therefore$ y-int $= 8$ 01

6 (b) $kx - y = 5$

$(3, 5)$

$k(3) - 5 = 5$

$3k = 5 + 5$

$\frac{3k}{3} = \frac{10}{3}$ 4 marks

$k = \frac{10}{3}$

7 (a) $c \propto m$

$c = km$

$k = \frac{c}{m}$

$k = \frac{15,000}{5}$

$k = 3,000$ 03

(b) $y \propto \frac{1}{x}$

(i) $y = \frac{k}{x^2}$

$k = yx^2 = 3 \times 4^2 =$

$= 3 \times 16$

$= 48$

$y = \frac{48}{x^2}$ 01

(ii) Find y when $x = 2$

$y = \frac{48}{2^2} = \frac{48}{4} = 12$ 01

$y = 12$

01

7 (c) Number of workers (w) varies inversely as the number of hours (h).

(i) $w \propto \frac{1}{h}$

$w = \frac{k}{h}$ 01

$w = 8$ workers

$h = 15$ hours

$8 = \frac{k}{15}$

$k = 120$

$w = \frac{120}{h}$ 01

(ii) 12 - workers, $h = ?$

$w = \frac{120}{h}$

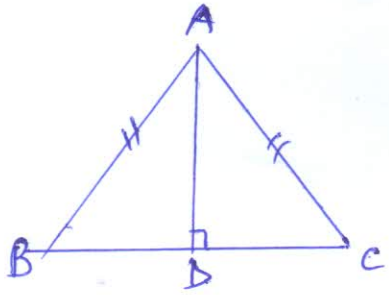
$12 = \frac{120}{h}$ 01

$12h = 120$

$h = \frac{120}{12} = 10$

∴ 12 workers needs 10 hrs 01

(a)



(i) $\overline{AB} = \overline{AC}$ (sides of an isosceles triangle)

$BD = \overline{DC}$ (D is the mid-point)

AD is common

$\triangle ABD \cong \triangle ACD$ (SSS postulate).

5 marks

(ii) $A\hat{B}D = A\hat{C}D$ (defn of congruence of triangles).

(b) $A\hat{B}C + A\hat{B}D = 180$ supp. angles

$A\hat{B}C + a = 180$

$A\hat{B}C = 180 - a$ ... (i)

$A\hat{C}E + A\hat{C}B = 180$ supp. angles

$A\hat{C}E + a = 180$

$A\hat{C}E = 180 - a$ ... (ii)

5 marks

$A\hat{B}C = A\hat{C}E = 180 - a$

hence $A\hat{B}C = A\hat{C}E$ (proved)

9 (a) x-int is 1 $\Rightarrow$ (1, 0)

y-int is 2 $\Rightarrow$ (0, 2)

gradient $= \frac{2-0}{0-1} = \frac{2}{-1} = -2$

$y = mx + c$

$y = -2x + 2$

04 marks

(b) $\frac{x}{2} + \frac{y}{2} = 1$

$x + y = 2$

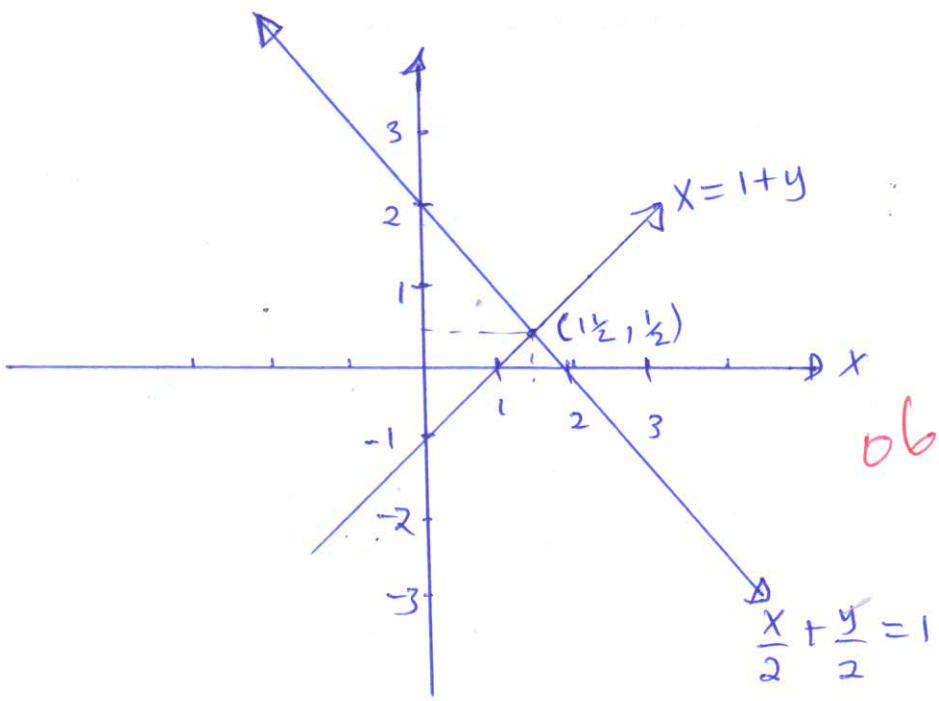
x-int $= 2$

y-int $= 2$

$x = 1 + y$

x-int $= 1$

y-int $= -1$



06 marks

Solution from graph $(x, y) = (1\frac{1}{2}, \frac{1}{2})$

10 a) Total cost

50 bags = 50 x 80,000

= 4,000,000/=

Add transport cost

4,000,000 + 200,000

Total cost = 4,200,000/= 03

(b) Total selling price per bag = 95,000

= 50 x 95,000

= 4,750,000 02

(c) Profit = Selling price − total cost

= 4,750,000 − 4,200,000

= 550,000 02

(d) Percentage profit

$$\% \text{ profit} = \frac{\text{Profit}}{\text{cost price}} \times 100$$

$$= \frac{550,000}{4200,000} \times 100$$

= 13.1 % 03